

Converging and diverging variations in metaphorization of light verbs: A corpus study on four Chinese speech communities

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Abstract

Metaphorization is a robust semantic process that underpins the development of grammatical categories and associated language variation and change. This study adopts a process-based approach to investigate variations in metaphorical extension of five common light verbs from different lexical sources in four Chinese speech communities: *jiyu* 給予 “GIVE”, *yu* 予 “GIVE”, *jia* 加 “ADD”, *gan* 幹 “WORK HARD”, and *nong* 弄 “FIDDLE”. Drawing on longitudinal (1995-2016) and latitudinal (Hong Kong, Taiwan, Beijing, Shanghai) data from the Pan-Chinese corpus LIVAC (https://en.wikipedia.org/wiki/LIVAC_Synchronous_Corpus) we demonstrate that light verbs develop differentially, ranging from uniform progression, significant variations, to even backward development. We also propose an innovative measure to quantify the distances in metaphorization between the regions and uncover that Chinese speech communities tend to converge on the process over time as led by Mainland Mandarin, despite clustering of Hong Kong and Taiwan and of Beijing and Shanghai, as also confirmed statistically. Based on the findings, we explore how the differential developments result from the interaction of underlying mechanisms of semantic change and lexical competition, as well as contact among Chinese varieties. Moreover, we highlight the importance of a corpus that rigorously monitors multiple regions in revealing covert variations in Global Chinese varieties.

1 Introduction

Language develops through changes such as enrichment of its lexical and grammatical repertoire. Thus there could be emergence of categorical shifts or new categories with semantic change (Heine et al., 1991; Hopper and Traugott, 2003, *i.a.*). Metaphors have been recognized to be instrumental in meaning extension from a concrete domain to an abstract domain (Traugott, 1982; Bybee and Pagliuca, 1985; Claudi and Heine, 1986; Sweetser, 1988). A salient case associated with robust metaphorization is **light verbs**, e.g., *give a person a slip*, where *give* loses the sense of ownership transfer and becomes semantically bleached, in contrast to the lexical “heavy” verb usage *give a person a book* (Jespersen, 1954, p.117ff; see also Cattell, 1984; Butt, 2010; Mohanan, 2017). Such **metaphorization** process is ubiquitous and can be found in the Germanic and Romance families (e.g., Alvarez-Morera, 2023; Pompei, 2023), Indo-Aryan languages (e.g., Hook, 1991; Butt and Geuder, 2003), Austronesian and Austroasiatic languages (e.g., Kwon, 2004; Nugraha, 2025), Sino-Tibetan languages including Tibetan (Lai, 2024) and Chinese (e.g., Diao, 2004; Tsou and Yip, 2020; Lu and Huang, 2023), among many others.

On the other hand, synchronic grammatical variations can reflect varying trajectories of diachronic grammaticalization processes (Weinreich et al., 1968; Brinton and Traugott, 2005). Nevertheless, little is known about the contribution of metaphorization to **language variations**. The issue is especially pressing in a language like Chinese, which lacks an inflectional morphology to signal grammaticalization and thus whose change and variation tend to be covert (Tsou

and Ji, 2022; Yip and Tsou, 2022; Tsou et al., 2023), in particular under the Global Chinese context (Tsou and You, 2003; Lin et al., 2019; W. C. J. Yip and Tang, 2022).

Light verbs have drawn the attention of Chinese linguists since the introduction of Modern linguistics (Lu, 1999/1980, p.294; Wang, 1985, p.142). Zhu, in his seminal work in 1985, referred to them as “bleached verbs” 虛化動詞, and identified 6 with notably different lexical sources: *jinxing* 進行 (from ENTER and WALK), *zuo* 作 (MAKE), *jiayi* 加以 (ADD), *geiyi* 給以, *jiyu* 給予 and *yuyi* 予以 (all from GIVE). Several works followed up on their grammatical properties (e.g., Zhou, 1987; Li and Chai, 1995; Yan, 1998; and many others). In 2004, corpus-based quantitative studies in different Pan-Chinese regions began to emerge. Diao, based on a self-curated corpus in Mainland China, extended the scope to *congshi* 從事 (WORK), as well as to monosyllabic *zuo* 做 (MAKE), *guo* 搞 (MAKE), *gan* 幹 (WORK HARD), *nong* 弄 (FIDDLE/PLAY), *jia* 加 (ADD), and *yu* 予 (GIVE). In the same year, Wang (2004) also drew corpus data from Taiwan to compare *zuo* 做, *guo* 搞, and *nong* 弄.

Areal differences were less noticed until Diao’s comparison in 2012 on four Pan-Chinese regions (Mainland China, Hong Kong, Taiwan, and Macau), concerning *jinxing* 進行 and *guo* 搞; and Huang et al.’s work in 2014 (later developed into Jiang et al., 2016, 2021; and Xu et al., 2022) on the variations in Mainland and Taiwan Mandarin, concerning *jinxing* 進行, *congshi* 從事, *zuo* 做, *guo* 搞, and *jiayi* 加以. Focusing on syntactic properties (e.g., aspectual markers and object types, etc.), they reached a common conclusion that light verbs in Mainland Mandarin are more grammaticalized than in Taiwan Mandarin. Diao additionally suggested a convergence of the four regions by observing borrowing of fixed expressions. Most recently, Tsou and Yip (2020) initiated comparison of metaphorization on a common but understudied light verb *da* 打 (from HIT) in Beijing, Hong Kong, and Taiwan with LIVAC, and similarly concluded that Beijing had a higher degree of metaphorical shift than Hong Kong and Taiwan. They also traced the longitudinal differences in metaphorization of

da-verbs and showed how the semantic process varies across the regions (see also Yip and Tsou, 2022 for extension to Macau). Lu and Huang (2023) and Kuo (2025) provided further insights that *yuyi* 予以 and *jiayi* 加以 are fully grammaticalized but *jiyu* 給予 is still half-way through the process, largely based on Taiwan data.¹ While there have been fruitful results on the latitudinal variations in grammaticalization and on the development trajectories of light verbs in Chinese,² less attention has been given to study metaphorization and its longitudinal variations (except Tsou and Yip, 2020), as well as wider implication. This calls for a large-scale quantitative study to explore whether different varieties of Chinese converge or diverge and the possible causation.

The current study proposes a novel **process-based** approach to investigate the metaphorization variations in Chinese. The empirical basis consists of five light verbs: *jiyu* 給予, *yu* 予, *jia* 加, *gan* 幹, and *nong* 弄. Unlike other light verbs that are fully metaphorized and no longer have the literal sense (e.g., *jinxing* 進行 “do” and *jiayi* 加以 “do/impose”), these five verbs are still undergoing metaphorization, as indicated by the polysemy between a literal meaning (lexical verb) and a metaphorical meaning (light verb), such as *jiyu shuben* 給予書本 “give books” and *jiyu zhichi* 給予支持 “give support”. They offer us a window to probe into the dynamic *process* of semantic bleaching resulting from metaphorization, over time and across regions.

Adopting such a dynamic perspective, we draw data in LIVAC from 1995-2016 among four Chinese speech communities: Hong Kong (HK), Taiwan (TW), Beijing (BJ), and Shanghai (SH) to compare the differential developments in their metaphorization processes. The key findings are:

- (i) Light verbs may develop differentially: while some have uniform progressions, some are still in flux and even develop “backwards”;
- (ii) There is a clear clustering between regions: the HK-TW cluster and the BJ-SH cluster;
- (iii) Despite the clustering, Mainland Mandarin has been leading the metaphorization process: (a) the light verbs are overall more metaphorized in BJ and SH; (b) other regions, especially TW, converge with BJ and SH in recent years.

¹ They classified these three light verbs as the GIVE-group despite the fact that *jiayi* 加以 comes from ADD.

² See also Cai et al. (2019) for comparison on “semantic prosody” between *jinxing* 進行 and *shaodao* 受到 in different registers; Jiang (2020) for the event semantics of light verbs.

	HK	TW	BJ	SH
<i>Jiyu</i> 給 予 (%)	6570 (22)	5202 (37)	<u>13099</u> (50)	6766 (39)
<i>Yu</i> 予 (%)	<u>9667</u> (32)	1567 (11)	714 (3)	435 (3)
<i>Jia</i> 加 (%)	<u>10876</u> (36)	<u>5132</u> (36)	4136 (16)	3872 (22)
<i>Gan</i> 幹 (%)	947 (3)	1015 (7)	<u>6894</u> (26)	3168 (18)
<i>Nong</i> 弄 (%)	2158 (7)	1206 (9)	1318 (5)	<u>3097</u> (18)
<i>Total</i> (%)	30218 (100)	14122 (100)	26161 (100)	17338 (100)

Table 1: Distribution of the five light verbs in 1995-2016 across regions

The rest of the paper is organized as follows. Section 2 introduces the corpus base LIVAC and the methodology for quantitative analyses. Section 3 overviews the polysemy and metaphorization of the five light verbs. Section 4 investigates the converging and diverging variations among the four speech communities. Section 5 concludes with remarks on the implications for light verb development and the importance of a rigorously curated corpus in large-scale measurement of variations in Global Chinese, as well as prospects.

2 Corpus base and methodology

Our data are drawn from the Pan-Chinese synchronous database LIVAC (Tsou and Kwong, 2015; <http://www.livac.org/>),³ featuring an innovative “Windows” approach that supports the process-based comparative perspective adopted here. Since 1995, LIVAC has regularly and rigorously cultivated over 750 million characters of representative media texts from Pan-Chinese communities including Beijing, Guangzhou, Hong Kong, Macau, Shanghai, Shenzhen, Singapore, and Taiwan, on a weekly basis and involving different subject domains. The curated data over three decades in LIVAC provide a unique basis for both latitudinal and longitudinal comparisons.

In total **87839** tokens of the five light verbs, *jiyu* 給予, *yu* 予, *jia* 加, *gan* 幹, and *nong* 弄, were found in LIVAC from 1995 to 2016 in HK, TW, BJ, and SH. A breakdown in Table 1 shows that the distribution is not even across regions. Boldface indicates the most common light verb(s) within a

region (two for TW as the difference is less than 0.5%), and underlining indicates the top region in which a light verb is most frequently used as compared to other regions. Some clustering already emerges: while HK and TW use *jia* 加 the most, BJ and SH (as well as TW) use *jiyu* 給予 the most.

All raw sentences were assigned a random number (using Excel’s =RAND()) and reordered. For each light verb, a maximum of 50 sentences were sampled from each year of two time periods: (a) 1995-2000, and (b) 2011-2016, from each region, giving **9781** sentences in total (i.e., over one tenth of total tokens). Both sampling and ordering are hence randomized, avoiding oversampling similar uses in consecutive sentences. Two annotators manually classified each token as literal (Type I), intermediate (Type II), or metaphorical (Type III) (criteria to be exemplified in Section 3), with disagreement adjudicated by a third annotator (i.e., each annotation is agreed upon by at least two annotators). To boost reliability, each rater attended a training session in which they were required to annotate a trial of 50 sentences with a follow-up work group meeting to resolve disagreement, before the actual annotation began. Krippendorff’s alpha was measured for inter-coder reliability using the `irr` package in R (with `ordinal` metric). The pre-adjudication agreement level is satisfactory ($\alpha = 0.82$), which is improved further after adjudication ($\alpha = 0.93$, counting the adjudicator and the closest rater). A sample annotation guideline is provided in Appendix A.

We follow Tsou and Yip (2020) and quantify the metaphorization degree of each light verb w within a given region r with the **Metaphorization Index (MI)** (0=fully literal; 100=fully metaphorized):

$$MI_r^w = \frac{\left(\frac{t_{II}^w}{2}\right) + t_{III}^w}{t_I^w + t_{II}^w + t_{III}^w}$$

($t_{I/II/III}^w$ =raw tokens of Type I, II, or III uses of w)

Variations in metaphorization can be measured by comparing the MI of a given light verb across regions, and changes can be measured by comparing the MI in different time spans. Note that the distribution of light verbs is uneven across regions, which might induce potential bias. MI avoids this by calculating the ratio without counting in frequency, allowing direct comparison between regions.

³ “Synchronous” means the corpus actively cultivates data every year and hence monitors development in real-time.

Nevertheless, discarding frequency entirely might create confounds for the overall pattern. Although we may aggregate the MI of all light verbs, two caveats follow: (i) while w_1 may have a higher MI in r_x than r_y , w_2 may well have a reversed distribution, a distinction that would disappear after direct aggregation; (ii) even if w_1 has the same MI in both regions, it might be far more frequent in one region over the other, which is not reflected by the MI itself.

To measure the magnitude of variations without masking data as such, we propose the **Metaphorization Distance Index (MDI)** below. It calculates the *distance* in metaphorization across regions. For ease of interpretability and avoiding confusion with MI, the MDI value is divided by 20 (hence, 0=closest; 10=furthest).

$$MDI_{r_y}^{r_x} = \frac{\sum_{i=1}^n |(t_{r_x}^{w_1} \times MI_{r_x}^{w_1}) - (t_{r_y}^{w_1} \times MI_{r_y}^{w_1})|}{20}$$

(r_x and r_y =two given regions; n =total number (*type*) of words; t_r^w and MI_r^w =normalized tokens and MI of a given word w in r respectively.)

The MDI sums the *differences* in the number of metaphorized tokens between two regions. The higher the MDI of r_x relative to r_y , the larger the difference between r_x and r_y . It is equipped with both word frequency and metaphorization degree to reflect synchronic variations in the above scenarios (i) and (ii). Longitudinal variations can be further measured by comparing the MDI values in different time periods.

Furthermore, we conducted **Ordinal Linear Regression (OLR)** by ORDINAL package in R (4.4.2), to model the degree of metaphorization (Type I/II/III) for each light verb. The model included *region* and *time* as fixed effects, along with their interaction, to examine whether regional variations and diachronic changes influenced the degree of metaphorization. Post hoc pairwise comparisons were performed to determine whether the latitudinal and longitudinal distributions of metaphorization types were statistically significant, with p-values adjusted using the Tukey method.

For a holistic analysis of the metaphorization process of all five light verbs in each region, we built a full model with three fixed effects from *region*, *time*, and *individual verbs* (Type~region+time+verb) to examine the variation due to these three factors and their contribution in the metaphorization process.

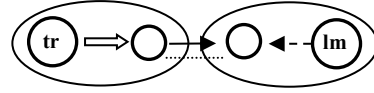


Figure 1: The image-schematic structure of TRANSFER (Langacker 2008, p.242)

3 Metaphorization of light verbs

Metaphorization, according to Heine et al. (1991, p.46), “involves a transfer, or a mapping of an image schema [in the sense of Sweetser (1988)], from one domain of conceptualization onto another”. It is a unidirectional process and proceeds from a concrete domain to an abstract one (Claudi and Heine, 1986), which results in generalization of meaning (Bybee and Pagliuca, 1985). During the course, semantic complexity may be lost with subsequent de-categorization, leading to development of a new category (Claudi and Heine, 1986). In other words, metaphorization is a process that leads to further semantic bleaching and grammaticalization.

The five light verbs under discussion are polysemous and have literal (**Type I**), intermediate (**Type II**) and metaphorical uses (**Type III**). The sample sentences extracted from LIVAC are given in the Appendix B. We propose that such polysemy is a result of metaphorization, and different senses share an image-schematic structure. One common schema is TRANSFER (Langacker, 1991, 2008; Halverson, 1999), which underpins the polysemy of GIVE verbs like *jiyu* 給予 and *yu* 予. In Figure 1, two participants are involved: the giver (trajector) and the recipient (landmark), with their own dominion (=ellipses). The trajector causes the object to move, such that the recipient gains subsequent access to it.

The literal uses of GIVE (I) involve a transfer of ownership of the object and corresponds to the TRANSFER schema, e.g., *jiyu shuben* 給予書本 “give books”. The object does not need to be tangible, e.g., *jiang tudi chanquan yizhuan yu taren* 將土地產權移轉予他人 “transfer the land property right to the others”. The intermediate uses (II) involve a causative meaning instead, e.g., *yu ren yixiang shenke* 予人印象深刻 “impress people”, where the impression is not “transferred” but rather caused by an individual to be perceived by another one, who is hardly counted as a recipient. Here, with conceptual metaphors (Lakoff and Johnson, 1980) like EMOTIONS ARE PHYSICAL OBJECTS, the causative uses can still be understood

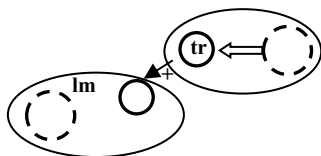


Figure 2: ADD's image schema



Figure 3 (left): WORD HARD's image schema

Figure 4 (right): FIDDLE's image schema

using the TRANSFER schema (Cervel, 2004). The same lexical items can thus extend their meaning from “transfer” to “cause”. The metaphorical uses (III) like *jiyu paichu* 給予排除 “to eliminate” and *zai-yu qianghua* 再予強化 “to strengthen (it) again” only involve transitive actions “elimination” and “strengthening”, and both GIVE verbs do not seem to contribute any concrete meaning. We regard these uses as genuine light verb usage, which, unlike Type I/II, can be replaced with other fully metaphorized light verbs *yuyi* 予以/*jiayi* 加以, and sometimes with *jinxing* 進行, but generally not the lexical verb *gei* 給 “give”. Again, with the metaphor ACTIONS ARE PHYSICAL OBJECTS, this use conforms to the TRANSFER schema where an action is imposed on the landmark by the trajector.

Jia 加 means addition of an object in its literal uses (I) like *jia shui* 加水 “add water” and *jia xi* 加戲 “add a scene”. We propose an image schema of ADD in Figure 2: the object (trajector) is added to another object's dominion (landmark), forming a part-whole relation. The causer of the addition is backgrounded (=dotted lines). Addition of an action/event is classified as intermediate uses (II), e.g., *jia kao* 加考 “have an additional exam” where *jia* 加 serves as a manner modifier. Likewise, this use is enabled by the metaphor ACTIONS ARE PHYSICAL OBJECTS. The metaphorical light verb uses (III) do not contribute lexical meaning and can be replaced by the disyllabic *jiayi* 加以, e.g., *da-jia zanshang* 大加讚賞 “to praise greatly”. This use falls under the image schema, where the causer (often foregrounded) “adds” a praise (via similar EMOTION/ACTION metaphors) to the target object that is understood as a person. The meaning of addition is also bleached, just like how the meaning of transfer is bleached for GIVE.

Gan 幹's literal sense (I) is WORK HARD, e.g., *gan zhonghuo* 幹重活 “do hard labor”. Its image schema is in Figure 3, where the trajector imposes

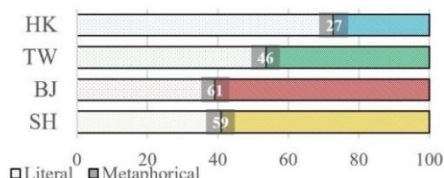


Figure 5: Aggregated MI of the four regions (95~16)

extra force on an underspecified, backgrounded object. The intermediate uses (II) are generalized to “work/engage in a job” with a foregrounded object and backgrounded effort, e.g., *gan baoan* 幹保安 “be a security guard” and *gan liang ge ban* 幹兩個班 “work two shifts”. They can often be replaced with *congshi* 從事/*zuo* 做. We also regard the sense of “conflict” as intermediate uses, where the hard effort remains foregrounded, e.g., *dui-zhe gan* 對著幹 “to work against (someone)”. The metaphorical uses (III) are fully generalized to “do” and can only be replaced with *zuo* 做, e.g., *gan shashi* 幹傻事 “do silly things” and *gan-xia duo qi xiji shijian* 幹下多起襲擊事件 “do/commit multiple attacks”. Here, the object in the schema is a foregrounded vehicle for the event/action. Note that only one participant is involved, unlike GIVE and ADD.

Nong 弄 literally means FIDDLE (often with hands) (I), e.g., *nong toufa* 弄頭髮 “fiddle hair”. Its image schema is in Figure 4, where the trajector causes an object (landmark) to move around but still within their own dominion. The intermediate uses (II) include “make” and “obtain”, e.g., *nong zaocan* 弄早餐 “make breakfast” and *nong qian* 弄錢 “earn/obtain money”, which we characterize as transformation of the landmark, either from an object A to B or from non-existence to existence. The metaphorical uses (III) include “do” (can be replaced with *gao* 搞), e.g., *nong shiqing* 弄事情 “do something” and *nong qingchu* 弄清楚 “make it clear”. Again, these uses are covered by the image schema and EMOTIONS/ACTIONS ARE PHYSICAL OBJECTS, with the loss of semantic complexity that the objects move around repeatedly.

Next, we demonstrate how metaphorization of these five light verbs develops differentially.

4 Variations in Chinese

The overarching observation is that BJ and SH are highly advanced in metaphorization, followed by TW, whereas HK is the least sensitive. Figure 5

	HK	TW	BJ	SH
HK	0.0	1.0	2.5	2.2
TW	1.0	0.0	2.3	1.9
BJ	2.5	2.3	0.0	1.3
SH	2.2	1.9	1.3	0.0

Table 2: The MDI matrix of the four regions (95~16)
(boldface=lowest value for each region)

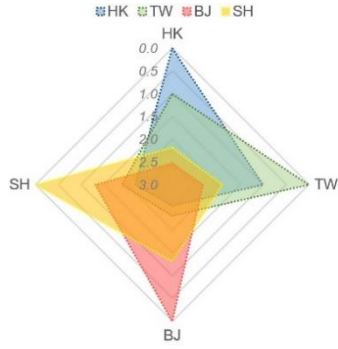


Figure 6: The radial graph of MDI of the four regions (95~16)

shows the aggregated MI in each region, i.e., the ratio of metaphorical tokens of all five light verbs. The results of the OLR model confirm the regional variations. BJ serves as the reference level for regional comparisons. Among the three regions, SH does not differ significantly from BJ ($B=-0.11$, $p=0.085$). In contrast, HK and TW are significantly different from BJ. HK shows the greatest divergence ($B=-0.98$, $p<0.001$), followed by TW, with a smaller but statistically meaningful divergence, and a lower degree ($B=-0.15$, $p=0.016$).

As we will reveal below, however, the development of the light verbs is not uniform.

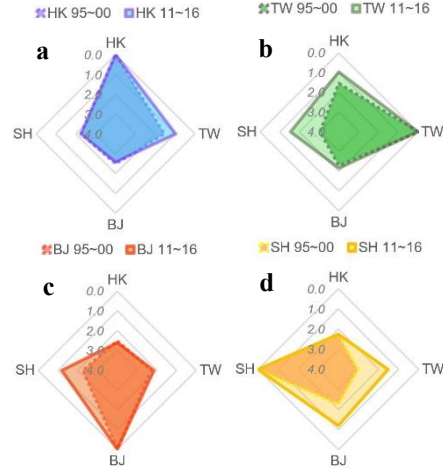
4.1 Convergence and divergence in the metaphorization process

First, clear **clustering** is obtained when measuring the distances between regions. The MDI of each pair of regions is given in Table 2, HK and TW are the closest to each other as indicated by the lowest MDI, so as BJ and SH. In contrast, HK-BJ is the most dissimilar pair, followed by BJ-TW and HK-SH. In other words, HK and TW converge together but diverge from BJ and SH in metaphorization. This is represented by the radial graph in Figure 6, where overlapping area indicates convergence and non-overlapping area indicates divergence.

Longitudinal comparison is achieved by subtracting the MDI value in 1995-2000 from that in 2011-2016, as shown in Table 3. Negative values signal a decrease in the distance, i.e., the two given regions have become more similar. Most values are

	HK	TW	BJ	SH
HK	0.0	-0.6	0.1	0.0
TW	-0.6	0.0	-0.1	-1.5
BJ	0.1	-0.1	0.0	-1.1
SH	0.0	-1.5	-1.1	0.0

Table 3: Change in MDI in the four regions (95~00 vs. 11~16) (boldface=lowest value for each region)



Figures 7a-d: Radial graphs of change in MDI of HK, TW, BJ, and SH (95~00 vs. 11~16)

negative, suggesting an overall convergence. The change in MDI is represented in Figures 7a-d for each region. Expansion of the light shaded area (from the dark one) indicates decreased distance (i.e., increased similarity) with other regions. HK tends to converge with TW whereas BJ with SH. On the other hand, TW also develops towards SH in addition to HK, and SH becomes more like both BJ and TW. Interesting, BJ does not converge towards TW, suggesting that SH's convergence with TW is independent of BJ's influence, which invites further explanation. Overall, we conclude that over time Chinese speech communities generally become more uniform in metaphorization.

To further observe the convergence/divergence of the light verbs in the four regions, we classified the light verbs into three groups: (A) in flux with significant variations; (B) uniform progression towards the final stage; (C) uniform incipient metaphorization with backward development.

4.2 Group A: Metaphorization in flux

The first group consists of GIVE light verbs *jiyu* 给予 and *yu* 予. As for *jiyu* 给予, all four regions are in the **mid-stage** of the process, with BJ and SH having the highest **MI** (both=**65**), followed by TW (**43**), and HK the lowest (**34**). *Yu* 予 shows a similar

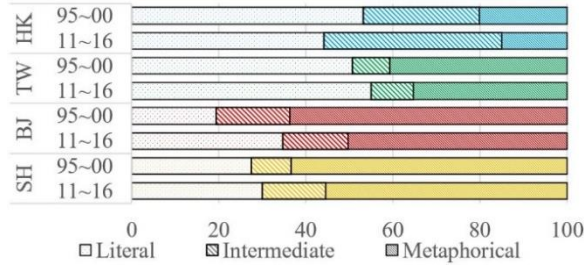


Figure 8: Variations in metaphorization of *jiyu* 給予

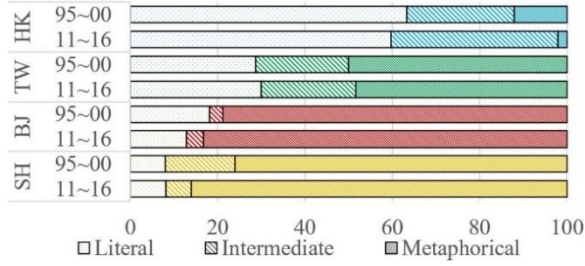


Figure 9: Variations in metaphorization of *yu* 予

pattern except that it is more metaphorized in SH (88), BJ (84), and TW (60) but not in HK (23). The metaphorization process is led by BJ and SH.

Figure 8 and Figure 9 show the breakdown of changes in each type of usage for both light verbs. HK shows an expansion of the intermediate usage. In addition, there is a complementary development in BJ and SH: they both have decreased metaphorical usage for *jiyu* 給予 yet increased metaphorical usage for *yu* 予. TW, on the other hand, remains relatively stable.

These observations are further confirmed by statistical analyses. BJ and SH show no significant difference in metaphorization for both *jiyu* 給予 ($B=-0.14$, $p=0.420$) and *yu* 予 ($B=-0.01$, $p=0.987$). However, HK and TW exhibit significantly lower odds of metaphorization for *jiyu* 給予 (HK: $B=-1.63$, $p<0.001$; TW: $B=-1.20$, $p<0.001$) and *yu* 予 (HK: $B=-3.04$, $p<0.001$; TW: $B=-1.24$, $p<0.001$).

4.3 Group B: Uniform progression

The second group consists of *gan* 幹 WORK HARD and *nong* 弄 FIDDLE. All the regions show uniform progression towards the **final stage**. For *gan* 幹, TW has the highest MI (81), followed by HK (76) and BJ (74), and lastly SH (69). *Nong* 弄 is more advanced overall, with HK having the highest MI (94), closely followed by SH (91), TW (88), and BJ (88). The breakdowns by usage are provided in Figure 10-Figure 11. BJ and SH share significant development of *gan* 幹 from intermediate usage to fully metaphorized usage, catching up on TW. HK shows a similar but weaker trend. *Nong* 弄 in BJ

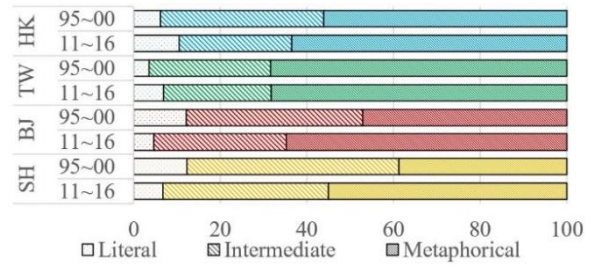


Figure 10: Variations in metaphorization of *gan* 幹

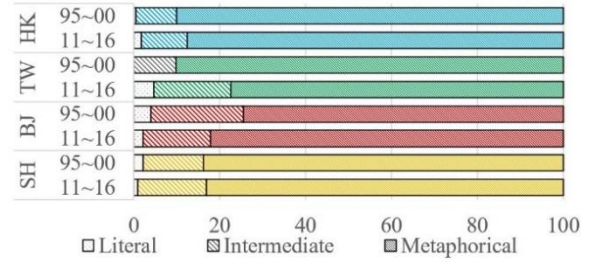


Figure 11: Variations in metaphorization of *nong* 弄

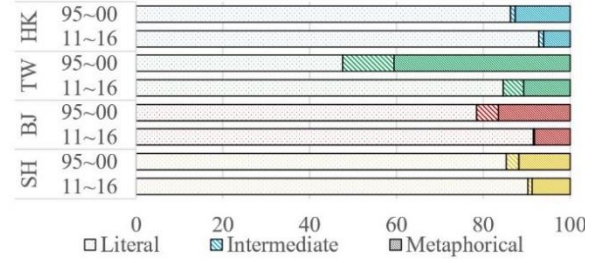


Figure 12: Variations in metaphorization of *jia* 加

has additional progression of metaphorization, contrasting the slight regression in TW.

Statistically, the pairwise comparison of regional variation supports the trend above. For *gan* 幹, BJ and SH show significant difference as compared to TW in 95~00, but in 11~16 only SH is significantly lower than TW in metaphorization ($B=-0.51$, $p=0.032$). As for *nong* 弄, in 95~00, BJ has significantly lower metaphorization degree than HK ($B=-1.14$, $p<0.001$), but it is no longer significant by 11~16 ($B=-0.42$, $p=0.314$). Instead, a meaningful difference emerges between HK and TW during the latter period ($B=-0.74$, $p=0.006$).

4.4 Group C: Incipient and backward development

The last group is a single light verb, *jia* 加 ADD. Unlike other light verbs, its metaphorization is in an **initial stage** in all four regions, as indicated by its lowest MI values: 29 in TW, 13 in BJ, 11 in SH, and 10 in HK. Strikingly, all the regions show **backward development**, as illustrated by the breakdown in Figure 12. TW undergoes drastic regression, “catching up” on HK, BJ, and SH.

The model results of time-wise comparison within each region also highlight the uniform

regression in metaphORIZATION of *jia* 加. There is a significant decline from 95~00 to 11~16 for BJ ($B=-0.986$, $p=0.0001$), HK ($B=-0.739$, $p=0.006$), and TW ($B=-1.731$, $p<0.001$). For SH, the decline of metaphORIZATION is slower with a marginal statistical significance ($B=-0.468$, $p=0.073$).

5 Concluding remarks

5.1 On the development of light verbs

One of our key findings is that light verbs are developed differentially, which may stem from some deeper mechanisms of categorial shifts as well as language variation and change. First, the grouping can be tied to the difference in their **image-schematic structures**. *Jiyu* 給予 and *yu* 予 share the TRANSFER schema and show parallel variations in metaphORIZATION (BJ/SH > TW > HK). On the other hand, *gan* 幹 and *nong* 弄, with similar uniform progression towards final stages, also both have a one-participant image schema, unlike TRANSFER with a second participant (recipient). Given that metaphoric extension is a systematic mapping between domains constrained by a semantic structure (e.g., Sweetser, 1988; Heine et al., 1991), light verbs with similar semantic structures are predicted to pattern together in diachronic development, which results in parallel synchronic variations. The grouping is corroborated by a previous insight that Chinese light verbs can be categorized into the GIVE-group and the DO-group (Jiang, 2020; Lu and Huang, 2023; Kuo, 2025). We suggest that the semantic underpinning of these two groups lies in whether a second participant is required in the underlying conceptual structures.

Second, some trajectories are attributable to **competition** among light verbs with similar semantic structures. *Yu* 予 is ahead of *jiyu* 給予 in metaphORIZATION. They even show complementary development in BJ and SH: progression for *yu* 予 but regression for *jiyu* 給予. The preference of *yu* 予 in expressing the metaphorical meaning is plausibly due to lexical competition and specialization of meaning to reduce the functional load of each lexical item (e.g., Clark, 1987). This perspective offers a potential explanation for the unexpected trends of *jia* 加, which has backward

development and contrasts the claim in Claudi and Heine (1986) that metaphORIZATION is unidirectional. While ADD and GIVE both have a two-participant schema, they differ in the lexical inventories in Chinese. GIVE is lexicalized as the morphemes *gei* 給 and *yu* 予, whereas ADD is only lexicalized as *jia* 加. Unlike *yu* 予/*jiyu* 給予, *gei* 給 remains a lexical verb and lacks the light verb usage (e.g., ungrammatical **zai-gei qianghua* 再給強化 vs. *zai-yu qianghua* 再予強化 “to strengthen (it) again”).⁴ It could well be that the lexical verb *gei* 給 “frees up” the room of metaphORIZATION for *yu* 予 and *jiyu* 給予, but *jia* 加 still bears the functional load to express the literal meaning, and hence the resistance to metaphORIZATION.⁵

Third, some convergence we witnessed can be traced to **contact** between speech communities. One such case is *yu* 予 in BJ and SH, which associate with similar socio-cultural backgrounds. We notice that the significant increase of *yu* 予’s Type III uses in SH (11~16) is largely due to popular usage of *yu jianwaizhihang* 予監外執行 “temporary releases (from prison)”. In LIVAC, this usage first appeared in BJ in 1999 (see Appendix B Example (6)) but was not found in SH from 95~00, which presumably entered SH in a later time. Another possible case is the advanced development of *gan* 幹 and *nong* 弄 in HK, which is unexpected given HK’s general insensitivity to metaphORIZATION (with the lowest MI), and low frequency of these two light verbs (see Table 1, in total only 10%). Both verbs are not commonly used in Cantonese, the predominant spoken language in HK, and hence are likely borrowed from other regions. If the time of borrowing is after the verbs well on the way to metaphORIZATION, the uniform progression in HK can be explained.

5.2 Large-scale dynamic measurement of variations with LIVAC, and prospects

Using an innovative measure, MDI, we revealed how speech communities **converge and diverge** in metaphORIZATION of light verbs, including the clustering of HK-TW and BJ-SH and the tendency of the four speech communities to converge as led by the Mainland. While the convergence has been suggested before (e.g., Diao, 2012a, 2012b), this

⁴ Interestingly, *gei* 給 enters another grammaticalization path to become a benefactive marker, e.g., *gei ta zuo dangao* 給他做蛋糕 “make a cake for him/her”.

⁵ How backward development arises might involve multiple factors and invites more future explanation.

study is the first of its kind to recruit a quantitative tool to measure **distances** between regions in light verb development. By comparing the two time periods (95~00 / 11~16), the measurement is bidimensional for tracking down how the trajectory of a dynamic process varies. Such a measurement can only be developed with a large-scale synchronous corpus like LIVAC, with both latitudinal and longitudinal dimensions from 30 years of texts in 6 major Chinese speech communities. Its rigorous “Windows” approach validates the comparison between regions and between time periods. LIVAC therefore provides an important basis for the study of grammatical and semantic processes in Chinese.

The characteristic absence of overt markings of grammatical categories in Chinese does not shield it from universal grammatical processes such as metaphorization. This is also reflected by other cases where words which could have different grammatical functions such as being a noun or a verb, e.g., *ai* 愛 “love” and 服務 “service/serve” (*ta fuwu shehui* 他服務社會 “s/he serves the society” vs. *ta wei shehui tigong fuwu* 他為社會提供服務 “s/he provides service to the society”). With a rigorously cultivated database, it can be shown that the preference in BJ for nominal uses, and hence more formal register, exceeds that in TW and HK (Kwong and Tsou, 2003). This contributes to the appreciation of **covert variations** within Chinese communities (Tsou and Ji, 2022) and deserves to be studied so as to better understand the Chinese language and its internal developments.

Achieving a better understanding of covert grammatical variations requires quantitative comparisons to measure the extent of a usage in a variety, which, under globalization, inevitably influences other varieties via cross-regional communication. A crucial task is to track the variations that may result from a change of feature-spreading from one region to another. This calls for a corpus **monitoring multiple regions**, which is uncommon in most existing ones. The current study showcases how a corpus with comparable cross-regional longitudinal data like LIVAC allows for systematic investigation of grammatical variation and change in **Global Chineses** (e.g., Tsou and You, 2003; Yip and Tsou, 2022).

5.3 Towards the future

Going forward, we plan to include more recent data after 2016 to closely monitor the development

of metaphorization, as enabled by the unique synchronous feature of LIVAC. We expect to reveal more intricate converging and diverging variations between Chinese varieties.

We also expect the current approach to illuminate studies in different mechanisms of metaphorization, for example, *via* a metaphorical agency in the object as for another light verb *da* (Kwong and Tsou, 2003; Tsou and Yip, 2020). Instead of direct metaphorical extension *via* image-schematic structures, *da*’s metaphorization relies on the type of its object, forming a continuum from concrete, quasi-concrete (as in *dazao qicheye de hangkongmujian* 打造汽車業的航空母艦 “(forge) an aircraft carrier of the automotive business”), to abstract nouns. How argument structures interact with metaphorization, and furthermore, how variations in argument structures (see Jiang et al. 2016, 2021 for variations in verb transitivity) correlate with variations in metaphorization, are yet other intriguing questions awaiting exploration.

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References

- Alvarez-Morera, Georgina. 2023. The nominal in light verb constructions: a corpus-based study in present-day English, German, Catalan and Spanish. Doctoral dissertation, Universitat Rovira i Virgili.
- Brinton, Laurel J., and Elizabeth C. Traugott. 2005. *Lexicalization and Language Change*. Cambridge: Cambridge University Press.
- Butt, Miriam, and Wilhelm Geuder. 2003. Light verbs in Urdu and grammaticalization. In Regine Eckardt, Klaus von Heusinger & Christoph Schwarze, eds., *Words in Time: Diachronic Semantics from Different Points of View*. 295–350. Berlin & New York: De Gruyter Mouton.
<https://doi.org/10.1515/9783110899979.295>
- Butt, Miriam. 2010. The light verb jungle: still hacking away. In Mengistu Amberber, Brett Baker and Mark Harvey, eds., *Complex Predicates: Cross-linguistic Perspectives on Event Structure*. 48–78. Cambridge, UK: Cambridge University Press.

- Bybee, Joan L., and William Pagliuca. 1985. Cross-linguistic comparison and the development of grammatical meaning. In Jacek Fisiak, ed., *Historical Semantics - Historical Word-Formation*. 59–83. Berlin & New York: De Gruyter Mouton. <https://doi.org/10.1515/9783110850178>
- Cai, Huiying, Yunhua Qu, and Zhiwei Feng. 2019. A corpus-based study of the semantic prosody of Chinese light verb pattern across registers: taking *jinxing* and *shoudao* as examples. *Glottometrics*, 46: 61–82.
- Cattell, Ray. 1984. *Composite Predicates in English*. North Ryde, New South Wales: Academic Press Australia.
- Clark, Eve V. 1987. *The Principle of Contrast: A Constraint on Language Acquisition*. Hillsdale, NJ: Erlbaum.
- Claudi, Ulrike, and Bernd Heine. 1986. On the metaphorical base of grammar. *Studies in Language*, 10(2): 297–335. <https://doi.org/10.1075/sl.10.2.03cla>
- Diao, Yanbin. 2004. *Xiandai Hanyu Xuyi Dongci Yanjiu* [A Study on Delexical Verb in Modern Chinese]. Dalian: Liaoning Normal University Press.
- Diao, Yanbin. 2012a. Lianganshidi xiandai hanyu changyongci “jinxing” shiyong qingkuang duibi kaoca yu fenxi [Comparison and analysis of the usage of “jinxing” in the four places across the Taiwan Strait]. *Journal of Wuling* 37(3): 116–124.
- Diao, Yanbin. 2012b. Tai-gang-ao diqu “gao” de shiyong qingkuang ji qi yu neidi de chayi [On the usage of *gao* in the regions of Taiwan, Hong Kong and Macao and the differences with the Mainland]. *Journal of Weinan Normal University*, 27(9): 80–88.
- Halverson, Sandra. 1999. Image schemas, metaphoric processes, and the “Translate” concept. *Metaphor and Symbol*, 14(3): 199–219. <https://doi.org/10.1207/S15327868MS140303>
- Heine, Bernd, Ulrike Claudi, and Friederike Hünemeyer. 1991. *Grammaticalization: A Conceptual Framework*. Chicago, IL: University of Chicago Press.
- Hook, Peter E. 1991. The emergence of perfective aspect in Indo-Aryan languages. In Elizabeth C. Traugott and Bernd Heine, eds., *Approaches to Grammaticalization 2*: 59–89. Amsterdam, Philadelphia: John Benjamins.
- Hopper, Paul J., and Elizabeth C. Traugott. 2003. *Grammaticalization* (2nd ed.). Cambridge: Cambridge University Press
- Huang, Chu-Ren, Jingxia Lin, Menghan Jiang, and Hongzhi Xu. 2014. Corpus-based study and identification of Mandarin Chinese light verb variations. In Marcos Zampieri, Liling Tan, Nikola Ljubešić and Jörg Tiedemann, eds, *Proceedings of the First Workshop on Applying NLP Tools to Similar Languages, Varieties and Dialects*, 1–10. Dublin: Association for Computational Linguistics and Dublin City University.
- Jespersen, Otto. 1954. *A Modern English Grammar on Historical Principles*. London: George Allen & Unwin & Copenhagen: Ejnar Munksgaard.
- Jiang, Haiyan. 2020. Chouxiang dongzuo shijian yuyi fanchou yu Hanyu xingshi dongci [Abstract action event semantic category and Chinese dummy verbs]. Doctoral dissertation, Jilin University.
- Jiang, Menghan, Dingxu Shi, and Chu-Ren Huang. 2016. Transitivity in light verb variations in Mandarin Chinese – a comparable corpus-based statistical approach. In *Proceedings of the 30th Pacific Asia Conference on Language, Information and Computation (PACLIC 30)*: 459–468. Seoul: Association for Computational Linguistics and Kyung Hee University.
- Jiang, Menghan, Hongzhi Xu, Jingxia Lin, Dingxu Shi, and Hunag Chu-Ren. 2021. Computational processing of varieties of Chinese: comparable corpus-driven approaches to light verb variation. In Marcos Zampieri and Preslav Nakov, eds., *Similar Languages, Varieties, and Dialects: A Computational Perspective*. 304–326. Cambridge: Cambridge University Press.
- Kuo, Pei-Jung. 2025. The GIVE group of the Mandarin light verbs. In Anna Riccio and Jens Fleischhauer, eds., *Light Verbs: Synchronic and Diachronic studies*. 43–64. Berlin & Boston: Düsseldorf University Press. <https://doi.org/10.1515/9783111388878-003>
- Kwon, Nayoung. 2004. A semantic and syntactic analysis of the causative structure in Vietnamese. In *Western Conference in Linguistics 2004 (WECOL 2004)*, Los Angeles, CA: University of Southern California.
- Kwong, Oi Yee, and Benjamin K. Tsou 2003. A synchronous corpus-based study of verb-noun fluidity in Chinese. In *Proceedings of the 17th Pacific Asia Conference on Language, Information and Computation*, 194–203. Tokyo: Association for Computational Linguistics and Waseda University.
- Lai, Ryan K. Y. 2024. Why we need asymmetric measures to classify multi-word expressions: the case of Tibetan light verb constructions. *Society for Computation in Linguistics* 7(1): 302–306. <https://doi.org/10.7275/scil.2211>
- Lakoff, George, and Mark Johnson. 1980. *Metaphor We Live By*. Chicago & London: University of Chicago.

- Langacker, Ronald. 1991. Cognitive Grammar. In Flip G. Droste and John E. Joseph, eds., *Linguistic Theory and Grammatical Description*. 275–306. Amsterdam & Philadelphia: John Benjamins Publishing Company
- Langacker, Ronald. 2008. *Cognitive Grammar: A Basic Introduction*. Oxford: Oxford University Press.
<https://doi.org/10.1093/acprof:oso/9780195331967.001.0001>
- Li, Feng, and Yun Chai. 1995. “V+yi” lei xuhua dongci de binyu [The objects of “V+yi” dummy verbs]. *Journal of Xinjiang Education Institute* 4: 68–71.
- Lin, Jingxia, Dingxu Shi, Menghan Jiang & Chu-Ren Huang. 2019. Variations in world Chineses. In Chu-Ren Huang, Zhuo Jing-Schmidt & Barbara Meisterernst, eds., *The Routledge Handbook of Chinese Applied Linguistics*, 196–211. London & New York: Routledge.
- Lu, Lu, and Chu-Ren Huang. 2023. A diachronic insight into the aspectual meaning in Light Verb Constructions. A case study in Mandarin Chinese. In Anna Pompei, Lunella Mereu and Valentina Piuino, eds., *Light Verb Constructions as Complex Verbs: Features, Typology and Function*. 305–336. Berlin & Boston: De Gruyter Mouton.
<https://doi.org/10.1515/9783110747997-012>
- Lü, Shuxiang. 1980. *Xiandai Hanyu Babaici* [Modern Chinese 800 Words]. Beijing: Commercial Press.
- Lü, Shuxiang. 1999. *Xiandai Hanyu Babaici (Zengdingben)* [Modern Chinese 800 Words (Enlarged Edition)]. Beijing: Commercial Press.
- Mohanan, Tara. 2017. Grammatical and light verbs. In Martin Everaert and Henk van Riemsdijk, eds., *The Wiley Blackwell Companion to Syntax* (2nd ed.). 1–27. Chichester, UK: John Wiley & Sons, Ltd.
- Nugraha, Danang S. 2025. A theoretical and corpus linguistics study of the light verb constructions: empirical data from Indonesian. Doctoral dissertation, University of Szeged.
- Peña Cervel, Sendra. 2004. The image-schematic basis of the event structure metaphor. *Annual Review of Cognitive Linguistics* 2: 127–158.
<https://doi.org/10.1075/arcl.2.05pen>
- Pompei, Anna. 2023. How light is ‘give’ as a Light Verb? A case study on the actionality of Latin Light Verb Constructions (with some references to Romance languages). In Anna Pompei, Lunella Mereu and Valentina Piuino, eds., *Light Verb Constructions as Complex Verbs: Features, Typology and Function*. 149–200. Berlin & Boston: De Gruyter Mouton.
<https://doi.org/10.1515/9783110747997-006>
- Sweetser, Eve E. 1988. Grammaticalization and semantic bleaching. In Shelly Axmaker, Annie Jaisser and Helen Singmaster, eds., *Proceedings of the 14th Annual Meeting of the Berkeley Linguistics Society*. Berkeley, CA: Berkeley Linguistics Society, 389–405.
<https://doi.org/10.3765/bls.v14i0.1774>
- Traugott, Elizabeth C. 1982. From propositional to textual and expressive meanings: some semantic-pragmatic aspects of grammaticalization. In Winfred P. Lehmann and Yakov Malkiel, eds., *Perspectives on Historical Linguistics*. 245–271. Amsterdam & Philadelphia: Benjamins.
- Tsou, Benjamin K. and Ji, Yaxuan. 2022. Yueyu he Xianhan zhong guanjian de yinxing chayi: yi jinyi fuhe ci wei li [Some salient covert differences between Cantonese and Modern Standard Chinese: near-synonymous compounds as examples]. Paper presented at The 26th International Conference on Yue Dialects, November 26–27, 2022, Jinan University.
- Tsou, Benjamin K., and Ka-Fai Yip. 2020. A corpus-based comparative study of light verbs in three Chinese speech communities. In *Proceedings of the 34th Pacific Asia Conference on Language, Information and Computation*. 302–311. Hanoi: Association for Computational Linguistics and VNU University of Science.
- Tsou, Benjamin K., and Oi Yee Kwong. 2015. LIVAC as a monitoring corpus for tracking trends beyond linguistics. *Journal of Chinese Linguistics Monograph Series* 25: 447–472.
- Tsou, Benjamin K., and Rujie You. 2003. *Huayu Yu Huaren Shehui* [Chinese Language and Society]. Hong Kong. City University of Hong Kong Press.
- Tsou, Benjamin K., Kelly Mak, and Kenny Mok. 2023. Towards the identification and tracking of salient traits and their developments in Chinese society via Big Data. *Advances in Techno-Humanities* 1: 4–18. Routledge.
<https://doi.org/10.4324/9781003376491-2>
- Wang, Leslie Fu-mei. 2004. A corpus-based study of mandarin verbs of doing. *Concentric: Studies in Linguistics* 30(1): 65–85.
- Wang, Li. 1985. *Zhongguo Xiandai Yufa* [Modern Chinese Grammar]. Beijing: Commercial Press.
- Weinreich, Uriel, William Labov, and Marvin Herzog. 1968. Empirical foundations for a theory of language change. In W. P. Lehmann and Y. Malkiel, eds., *Directions for Historical Linguistics*, 97–195. Austin, TX: University of Texas Press.
- Xu, Hongzhi, Menghan Jiang, Jingxia Lin, and Chu-Ren. 2022. Light verb variations and varieties of Mandarin Chinese: Comparable corpus driven

approaches to grammatical variations. *Corpus Linguistics and Linguistic Theory*, 18(1): 145–173. <https://doi.org/10.1515/cllt-2019-0049>

Yan, Zhongsheng. 1998. Shuo “houxu dongcixing binyu dongci” [On “verbs taking verbal objects”]. *Journal of Hebei Normal University (Natural Science)* 2: 91-93.

Yip, Ka-Fai, and Benjamin K. Tsou. 2022. MSC variations in metaphorization among Pan-Chinese speech communities. Paper presented at the 28th Annual Conference of the International Association of Chinese Linguistics, May 20-22, 2022, The Chinese University of Hong Kong.

Yip, Wai Chi J., and Sze-Wing Tang. 2022. Lexical variations in Asian Chinese speaking communities: a corpus-informed study of online, offline, and digital. *Global Chinese*, 8(2): 161–187. <https://doi.org/10.1515/glochi-2022-0001>

Zhou, Gang. 1987. Xingshi dongci de cifenlei [Subdivision of dummy verbs]. *Chinese Language Learning* 1: 11–14.

Zhu, Dexi. 1985. Xiandai shumian hanyu li de xuhua dongci he mingdongci [Dummy verbs and nominal verbs in Modern Written Chinese]. *Journal of Peking University (Humanities and Social Sciences)* 5: 1–6.

A Sample annotation guidelines for *yu* 予

(only instructions translated)

Type I

- General meaning: Transfer ownership
- Object type: usually concrete nouns
- Auxiliary means for classification: Be able to identify and/or recover a **giver/subject**, a **recipient/indirect object**, and a **patient/direct object**.
- Examples:
 - //外交部應會分階段，分國家，分地區、分時期，漸進◆予東協 8 國及印度免簽、落地簽或電子簽//
 - No explicit patient: //甚至由法官/檢察官口述◆予書記官記錄//
 - No explicit recipient or patient: //說得嚴苛點是「天◆予不取、反受其咎」。//
 - No explicit give or recipient: //將◆予 5-10 萬元（人民幣）的獎勵//
 - Property rights and power's (temporary) transfers count as Type I, e.g., //授權◆予董事會通過// and //始得將土地產權移轉◆予他人//

- Information transfer (via any media) count as Type I, e.g., //傳短訊◆予胞弟道別//

Type II

- General meaning: Causative/ let/ allow
- Object type: usually abstract nouns, as in: 「予…的感覺」“give ... feeling”、 「予…的印象」“give ... impression”、 「予人口實」“give one's critics a handle”、 「予人方便」“give ... convenience”, etc.
- Auxiliary means for classification: usually can be replaced with *gei* 給 “give”
- Examples:
 - //他不願◆予人以「鷹派」的印象//
 - //甚至◆予人有草率之譏//
 - //同時也跨海放映該會「重生」紀錄片◆予北京清大師生觀賞。//

Type III

- General meaning: Do/ implement an action
- Object type: usually eventive noun (including verb noun), as in: 「予支持」“support”、 「予懲罰」“punish”、 「予排除」“eliminate”, etc.
- Auxiliary means for classification: *cannot* be replaced with *gei* 給 “give”
- Examples:
 - //最近星國政府正打算修法再◆予強化//
 - //均◆予不起訴。//

Borderline cases

- ~建議 “suggestion”: Type I (information)
- ~評級/評價 “rate”: Type I
- ~榮譽 “honour”: Type I
- ~時間 “time”: Type II
- ~機會 “opportunity” Type II
- ~壓力 “stress”: Type II
- ~回答 “answer”: Type II
- ~保障 “protection”: Type II
- ~自治 “autonomy”, ~獨立 “independence”: Type II
- ~豁免 “exemption”: Type III
- ~安排 “arrangement”: Type III

- ~肯定 “affirmation”, ~關心 “care”: Type III
- ~治療 “treatment”: Type III
- ~指引 “guidelines”: if the context implies actual guidelines (e.g., by distribution), count as Type I; otherwise Type III
- ~援助 “aid”: if the context implies money or supplies, count as Type I; otherwise Type III

B Sample sentences of the five light verbs from LIVAC

The polysemy of *jiyu* 給予

- (1) Type I: ownership transfer
Wunv jianzhuang, jingkong xia quchu siqian yibai yuan jiyu zeiren 吳女見狀，驚恐下取出四千一百元給予賊人 “Noticing the situation, Ms. Ng gives the robber \$4100 in fear.” (HK 1997)
- (2) Type II: causative
Jiji zhengqu guojia jiyu benshi xianshixianhang de teshu zhengce 積極爭取國家給予本市先試先行的特殊政策 “(Our city) strives for the “try first” special policy offered by the central government.” (SH 2013)
- (3) Type III: light verb DO, “impose”
Yifa yigui jinxing jiemian yuetan, tongbao piping shenzhi jiyu xingzheng chufen 依法依規進行誡勉約談、通報批評甚至給予行政處分 “(The new government) legally performs exhortations, condemnations and even presents administrative punishments.” (BJ 2014)

The polysemy of *yu* 予

- (4) Type I: ownership transfer
Guotai churang guquan yu Zhonxintaifu duonian 國泰出讓股權予中信泰富多年 “Cathay Pacific has been transferring their shares to CITIC Pacific for many years.” (TW 1997)
- (5) Type II: causative
Qizhong you liang-wei zhuangaoren yu wo yinxiang shenke 其中有兩位撰稿人予我印

象深刻 “Among them, two writers impressed me” (TW 2012)

- (6) Type III: light verb DO, “impose”
Dui youguan bumen banli jianxing, jiashi, zan-yu jianwaizhihang, baowaijiuyi zhong de weifaqing kuang, tichu jiuzheng yijian 9672 jianci 對有關部門辦理減刑、假釋、暫予監外執行、保外就醫中的違法情況，提出糾正意見 9672 件次 “According to the illegal situations in relative agencies’ commutation of sentences, paroles, temporary releases and compassionate releases, rectifications are declared 9672 times.” (BJ 1999)

The polysemy of *jia* 加

- (7) Type I: addition of objects
Jingye de shenjianhong shuo, ru jia zhe chang xi neng rang juqing geng you xiaoguo, mei wenti 敬業的沈建宏說，如加這場戲能讓劇情更有效果，沒問題 “The dedicated Shen says, ‘No problem if adding this scene makes the plot more impressive.’” (TW 2015)
- (8) Type II: addition of events (manner)
Zai bixuankao de yi men kemu zhiwai, zai yaoqiu jia-kao yi zhi liang ge kemu 在必選考的一門科目之外，再要求加考一至兩個科目 “Apart from the single compulsory subject, one or two subject(s) is/are required to be additionally taken.” (SH 1998)
- (9) Type III: light verb DO, “impose”
Bushaoren dui qiaobusi da-jia zanshang 不少人對喬布斯大加讚賞 “Many people greatly praise Jobs.” (BJ 2012)

The polysemy of *gan* 幹

- (10) Type I: “work hard”
Quancun nannv laoyou da-gan ku-gan 全村男女老幼大幹苦幹 “The whole village, men and women, young and old, all worked hard through the tough jobs.” (BJ 1995)
- (11) Type II: “work/engage in a job”
Cong biao mian shang kan gan de shi guoji zixun yewu zhe yi hang 從表面上看幹的是國際諮詢業務這一行 “It looked like they were doing the international consulting business from the surface.” (SH 1996)

(12)Type II: “conflict”

Buguo Yashi xinwenbu que andili “dui-zhe gan”
不過亞視新聞部卻暗地裏「對著幹」 “But
ATV news department was working against
them secretly.” (HK 2011)

(13)Type III: Light verb DO

Ni haipa wu-gan shashi 你害怕誤幹傻事
“You’re afraid of doing something stupid by
mistake.” (TW 2015)

(14)Type III: Light verb DO, “commit (crimes)”

*Youde huangniu luzi geng wai, zhijie gan-qi-le
jiapiao goudang* 有的黃牛路子更歪，直接
幹起了假票勾當 “Some ticket scalper even
took a dirtier route and went straight into
selling fake tickets.” (SH 2014)

The polysemy of *nong* 弄

(15)Type I: “fiddle, play” (often with hands)

*Chenrongsen geng chen Keyi nong toufa zhiji,
liuzhidaji* 陳容森更趁可頤弄頭髮之際，溜
之大吉 “Chan Yung Sam quietly ran away
while Ho Yee was fixing her hair.” (HK 2000)

(16)Type II: creation, “make”

*huidao jia hai de wei qinv he liang ge gege
nong chide* 回到家還得為妻女和兩個哥哥
弄吃的 “He still had to prepare food for his
wife, daughter, and two brother when he got
home.” (SH 1996)

(17)Type II: “obtain, get”

*Qiudui de mubiao shi “zhan dian xianqi, tian
dian baqi, nong dian yunqi, xue dian caiqi”* 球
隊的目標是「沾點仙氣，添點霸氣，弄點
運氣，學點才氣」 “The team’s goal is to
“Catch some magic, add some aura, get some
luck, and have some talent” (SH 2013)

(18)Type III: light verb DO

Laoyang, zhe ge shi ni kan zenme nong a? 老
楊，這個事你看怎麼弄啊？ “Old Yang, how
should we do about this?” (BJ 2013)

(19)Type III: light verb CAUSE

*bushao guanzhong bei ta paianjiaojue de
fangshi nong-de xinyangyang* 不少觀眾被他
拍案叫絕的方式弄得心癢癢 “Many viewers
got excited by his amazing selling style.” (TW
2016)